

B) 25%.



C) 40%.



Explanation

Coefficient of variation, $CV = \text{standard deviation} / \text{mean}$. The standard deviation is the square root of the variance, or $4^{1/2} = 2$. So, $CV = 2 / 10 = 20\%$.

(Module 3.1, LOS 3.b)

Question #31 of 61

Question ID: 1572764

A distribution that has positive excess kurtosis is:

A) more peaked than a normal distribution.



B) more skewed than a normal distribution.



C) less peaked than a normal distribution.



Explanation

A distribution with positive excess kurtosis is more peaked and has fatter tails than a normal distribution.

(Module 3.2, LOS 3.c)

Question #32 of 61

Question ID: 1572714

An investor has a portfolio with 10% cash, 30% bonds, and 60% stock. If last year's return on cash was 2.0%, the return on bonds was 9.5%, and the return on stock was 25%, what was the return on the investor's portfolio?

A) 12.17%.



B) 18.05%.



C) 11.77%.



Explanation

Find the weighted mean of the returns. $(0.10 \times 0.02) + (0.30 \times 0.095) + (0.60 \times 0.25) = 18.05\%$

Asset	Weight	Return	Weight × Return
Cash	10%	2%	$10\% \times 2\% = 0.2\%$
Bonds	30%	9.5%	$30\% \times 9.5\% = 2.85\%$
Stock	60%	25%	$60\% \times 25\% = 15\%$
Weighted Average Return $\Sigma \text{ Weight} \times \text{Probability}$			18.05%

(Module 3.1, LOS 3.a)

Question #33 of 61

Question ID: 1572723

An investor has a \$12,000 portfolio consisting of \$7,000 in stock P with an expected return of 20% and \$5,000 in stock Q with an expected return of 10%. What is the investor's expected return on the portfolio?

A) 15.8%.



B) 15.0%.



C) 30.0%.



Explanation

Here we need to multiply the returns by the proportion that each stock represents in the portfolio then sum.

Stock	Return	Invested	Proportion of Portfolio	Return × Proportion
P	20%	\$7,000	7/12	$20\% \times 7/12$
Q	10%	\$5,000	5/12	$10\% \times 5/12$
Total		\$12,000		15.83%

(Module 3.1, LOS 3.a)

Question #34 of 61

Question ID: 1572722

Michael Philizaire decides to calculate the geometric average of the appreciation/depreciation of his home over the last five years. Using comparable sales and market data he obtains from a local real estate appraiser, Philizaire calculates the year-to-year percentage change in the value of his home as follows: 20, 15, 0, -5, -5. The geometric return is *closest* to:

A) 0.00%.



B) 4.49%.



C) 11.60%.



Explanation

The geometric return is calculated as follows:

$$[(1 + 0.20) \times (1 + 0.15) \times (1 + 0.0) (1 - 0.05) (1 - 0.05)]^{1/5} - 1,$$

$$\text{or } [1.20 \times 1.15 \times 1.0 \times 0.95 \times 0.95]^{0.2} - 1 = 0.449, \text{ or } \mathbf{4.49\%}.$$

(Module 3.1, LOS 3.a)

Question #35 of 61

Question ID: 1572717

The respective arithmetic mean and geometric mean returns of the following series of stock market returns are:

Year 1	14%
Year 2	6%
Year 3	-5%
Year 4	20%

A) 8.75%; 8.62%.



B) 8.90%; 8.62%.



C) 8.75%; 8.34%.



Explanation

$$(14 + 6 + (-5) + 20) / 4 = 8.75.$$

$$((1.14 \times 1.06 \times 0.95 \times 1.20)^{0.25} - 1 = 8.34\%.$$

(Module 3.1, LOS 3.a)

Question #36 of 61

Question ID: 1572769

Which of the following statements concerning kurtosis is *most* accurate?

- A) A distribution with kurtosis of +2 has fatter tails than a normal distribution. 
- B) A leptokurtic distribution has excess kurtosis less than zero. 
- C) A leptokurtic distribution has fatter tails than a normal distribution. 

Explanation

A leptokurtic distribution is more peaked than normal and has fatter tails. However, the excess kurtosis is greater than zero.

(Module 3.2, LOS 3.c)

Question #37 of 61

Question ID: 1572739

What are the median and the third quintile of the following data points, respectively?

9.2%, 10.1%, 11.5%, 11.9%, 12.2%, 12.8%, 13.1%, 13.6%, 13.9%, 14.2%, 14.8%,
14.9%, 15.4%

- A) 12.8%; 13.6%. 
- B) 13.1%; 13.6%. 
- C) 13.1%; 13.7%. 

Explanation

The median is the midpoint of the data points. In this case there are 13 data points and the midpoint is the 7th term.

The formula for determining quantiles is: $L_y = (n + 1)(y) / (100)$. Here, we are looking for the third quintile (60% of the observations lie below) and the formula is: $(14)(60) / (100) = 8.4$.

The third quintile falls between 13.6% and 13.9%, the 8th and 9th numbers from the left. Since L is not a whole number, we interpolate as: $0.136 + (0.40)(0.139 - 0.136) = 0.1372$, or 13.7%.

(Module 3.1, LOS 3.a)

Question #38 of 61

Question ID: 1572733

An analyst compiles the returns on Fund Q over the last four years:

Year	Return
1	4%
2	3%
3	2%
4	30%

Which of the following will result in the *lowest* measure of the mean return?

- A) The arithmetic mean.

✖
- B) The geometric mean.

✖
- C) The harmonic mean.

✔

Explanation

Harmonic mean = $\frac{4}{\frac{1}{1.04} + \frac{1}{1.03} + \frac{1}{1.02} + \frac{1}{1.30}} - 1 = 0.0864 = 8.64\%$

Geometric mean = $[(1.04)(1.03)(1.02)(1.30)]^{\frac{1}{4}} - 1 = 0.0917 = 9.17\%$

Arithmetic mean = $\frac{4\% + 3\% + 2\% + 30\%}{4} = 9.75\%$

(Module 3.1, LOS 3.a)

Question #39 of 61

Question ID: 1572746

Annual Returns on ABC Mutual Fund									
Yr 1	Yr 2	Yr 3	Yr 4	Yr 5	Yr 6	Yr 7	Yr 8	Yr 9	Yr 10
11.0%	12.5%	8.0%	9.0%	13.0%	7.0%	15.0%	2.0%	-16.5%	11.0%

Assuming a mean of 7.2%, what is the sample standard deviation of the returns for ABC Mutual Fund for the period from Year 1 to Year 10?

- A) 7.8%.

✖
- B) 9.8%.

✖
- C) 9.1%.

✔

Explanation